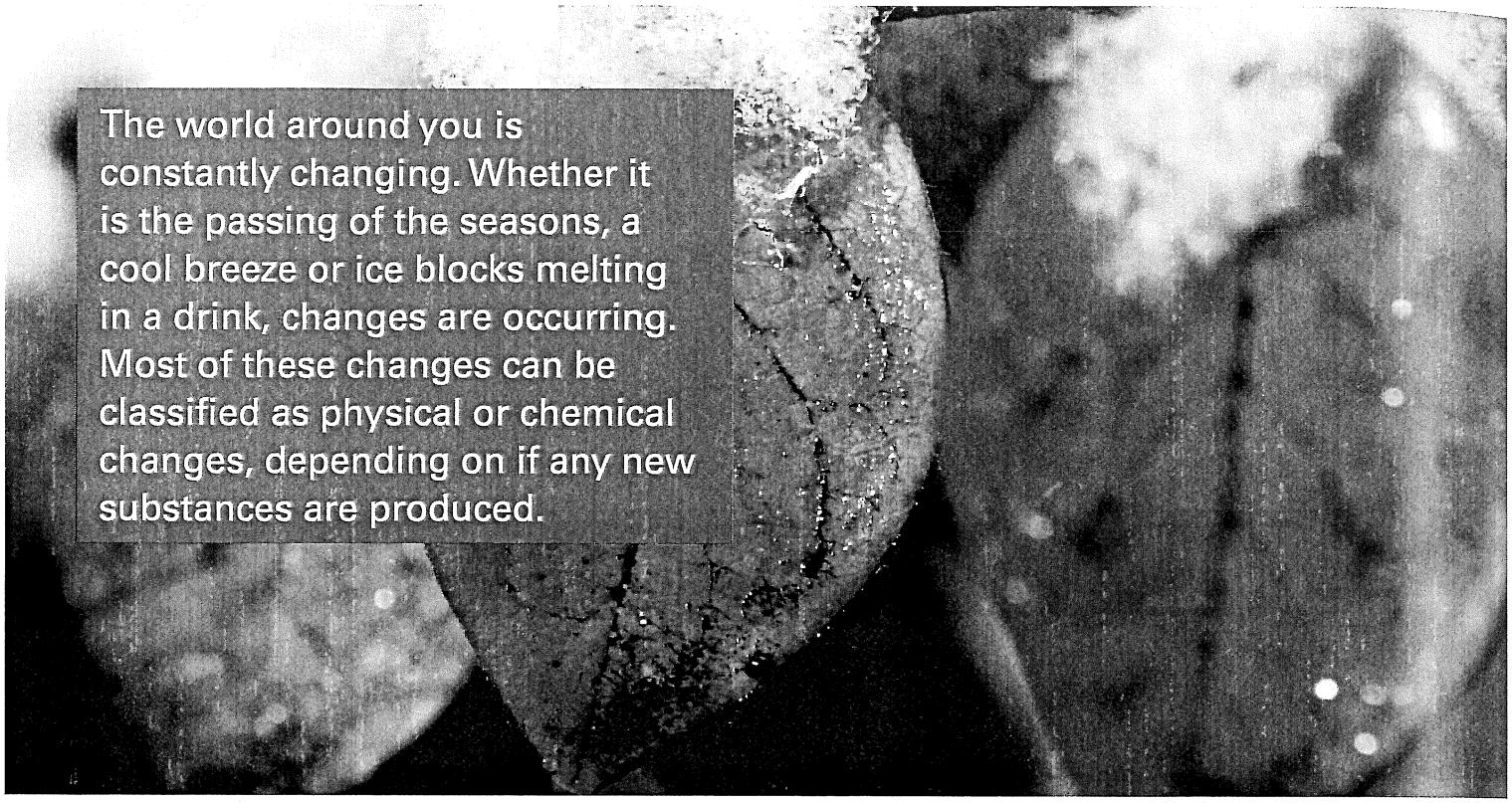
MODULE
7.1 When substances change



The world around you is constantly changing. Whether it is the passing of the seasons, a cool breeze or ice blocks melting in a drink, changes are occurring. Most of these changes can be classified as physical or chemical changes, depending on if any new substances are produced.

science 4 fun

Invisible ink



Can you use a chemical change to write a secret message?

Collect this ...

- lemon juice
- paper
- fine paintbrush
- warm light globe or hairdryer

Do this ...

1. Use the paintbrush to write a secret message on the paper in lemon juice.
2. Let the paper dry and then give the secret message to a friend.
3. To read the message, tell your friend to hold the message up to a warm light globe or use a hairdryer to heat the paper.

Record this ...

1. Describe what happened.
2. Explain why you think this happened.

Identifying physical change

The key feature of a **physical change** is that no new substance is produced during the change. A substance undergoes a physical change if it changes shape or form, if it expands (gets bigger) or contracts (gets smaller), if it changes state (into a solid, liquid or gas), or if it mixes with another substance. Figure 7.1.1 shows some examples of physical changes. The **physical properties** of the substance may change but the substance before and after the change is exactly the same. For this reason, physical changes can often be reversed.

Changing shape or form

The simplest example of a physical change is a change in the shape or form of a substance. In this case, a force is applied to break, bend, stretch, crush, twist or compress the object. These changes do not produce new substances. For example, when you squash an aluminium can, its shape changes but the can is still made of aluminium. Similarly, if you crush an aspirin tablet into a powder, it is still aspirin but in a different form. These changes can be reversed relatively easily.

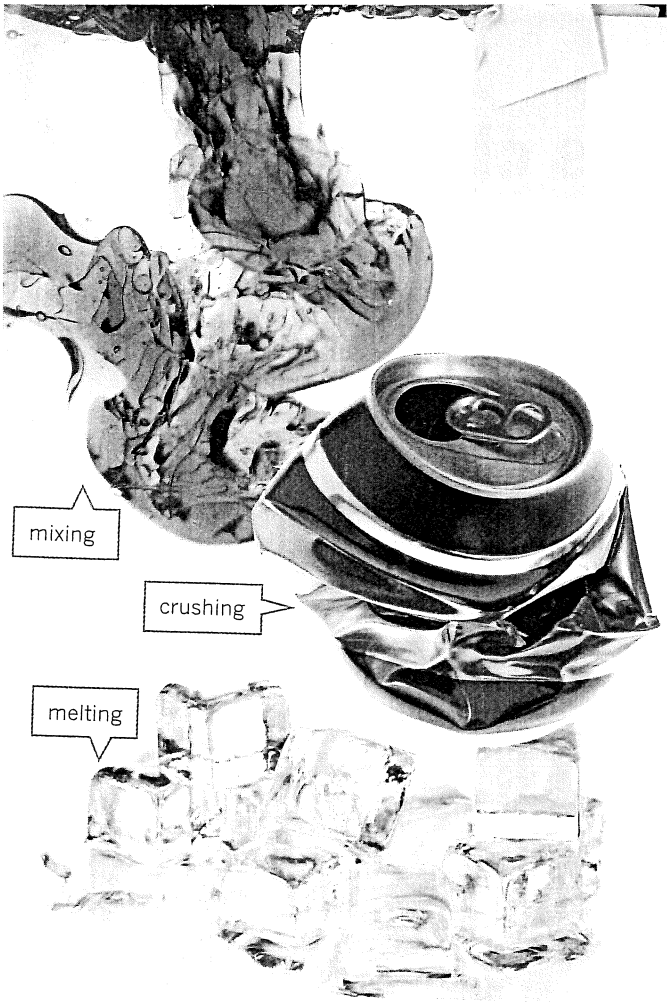


FIGURE 7.1.1 During these physical changes, no new substances are produced.

If you bend a metal rod in one direction then you can easily bend the rod back to its original shape. Even the crushed aspirin powder can be compacted back into the form of an aspirin tablet.

Expansion and contraction

Physical changes also occur when the temperature of a substance increases or decreases. When solids, liquids and gases are heated, they expand and take up more space. When **expansion** occurs, the volume of the object increases and its density decreases, but no new substances are formed. The reverse of expansion is **contraction**. When a solid, liquid or gas is cooled, it contracts—the volume of the substance decreases and its density increases.

Thermometers use expansion and contraction of liquids such as ethanol and mercury to measure temperature. When the temperature increases, the liquid in the thermometer expands and moves up the thermometer. When the temperature decreases, the liquid contracts and moves down.

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Hot-air balloons use the expansion of heated air to lift off the ground. Figure 7.1.2 shows how hot-air balloons use a burner to heat the air inside. As the air heats, it expands to fill the balloon. The hot air inside is much less dense than the cooler air outside, and so the balloon floats and lifts off the ground. The hot air in the balloon takes up more space but is still the same substance as it was before it was heated. Therefore, expansion is a physical change.

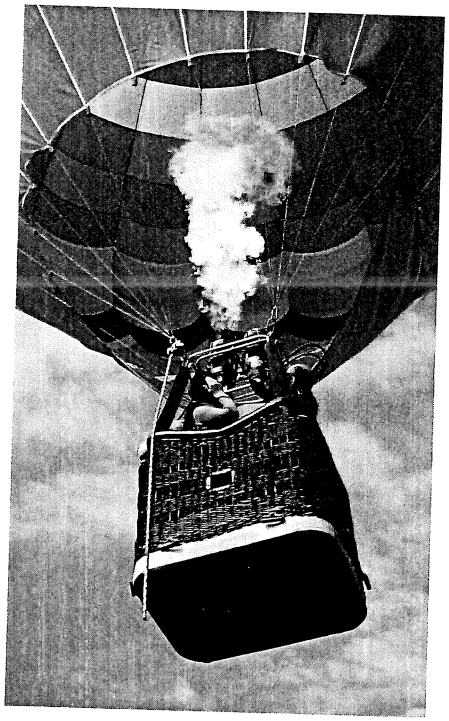


FIGURE 7.1.2 Heating the air inside the balloon causes it to expand and become less dense than the air outside. As a result, the balloon floats.

Changes of state

If enough heat is applied to a substance, then it will change state from a solid to a liquid (**melting**) or from a liquid to a gas (**evaporation**). These changes are known as changes of state. When a substance is cooled, the change of state is reversed from a gas to a liquid (**condensation**) or from a liquid to a solid (**freezing** or **solidification**). A few substances change directly from a solid to a gas (**sublimation**) or from a gas to a solid (**deposition**).

Melting

When a solid is heated enough, it will melt and change into a liquid. For example, when you place an ice cube in your mouth, the heat from your mouth melts it into liquid water. Likewise, when butter is spread on hot toast, the butter melts and soaks into the bread. The physical properties of the solid and the liquid may be very different, but they are both made from the same substance. Therefore melting is a physical change.

Freezing

When a liquid is cooled enough, it will form a solid. This is what happens when liquid water is put in the freezer to form ice cubes. Freezing can also be seen when drops of liquid candle wax cool and become solid again. Freezing converts liquids into solids, so is the reverse of melting.

Evaporation

Evaporation occurs when a liquid is warmed and becomes a gas. Evaporation occurs at the surface of a liquid at all temperatures, but it becomes faster as the temperature of the liquid increases. This is the reason that puddles of water dry faster on a hot day. It is also why tumble-driers heat clothes to dry them more quickly. If the liquid is heated above a certain temperature, known as the boiling point, then bubbles of gas will form within the liquid and the liquid boils.

Condensation

The opposite of evaporation is condensation. Condensation is when a gas cools and becomes a liquid. When you leave a cold can of soft drink out of the fridge, it develops moisture on the outside, as shown in Figure 7.1.3. This is because water vapour in the air condenses on the cold outside of the can to form liquid water. Similarly, when you breathe out on a cold day, 'fog' may come out of your mouth. This 'fog' is water vapour in your breath condensing to form a mist of tiny liquid water droplets.

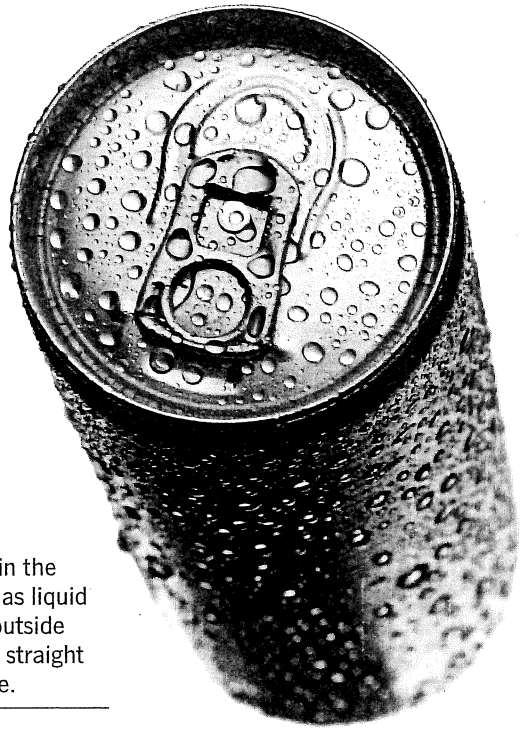


FIGURE 7.1.3 Water vapour in the air condenses as liquid water on the outside of a can taken straight from the fridge.

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Water out of thin air

Engineers at the University of Engineering and Technology (UTEC) in Lima, Peru, have created a billboard that supplies pure drinking water. The area has rainfall of less than 1.3 cm per year, so locals rarely have enough clean water to drink. However, humidity is very high. The billboard uses special filters to condense the water vapour out of the air. In this way, it can produce more than 3000 litres of water per month. This is enough to supply hundreds of families with fresh water.



FIGURE 7.1.4 A local Peruvian boy drinks fresh water collected from the UTEC billboard.

Mixing

The mixing of two substances is another type of physical change. Figure 7.1.5 shows a bucket of different-coloured balls. It is an example of a mixture that can be seen by eye. The different-coloured balls mix but no new balls are formed.

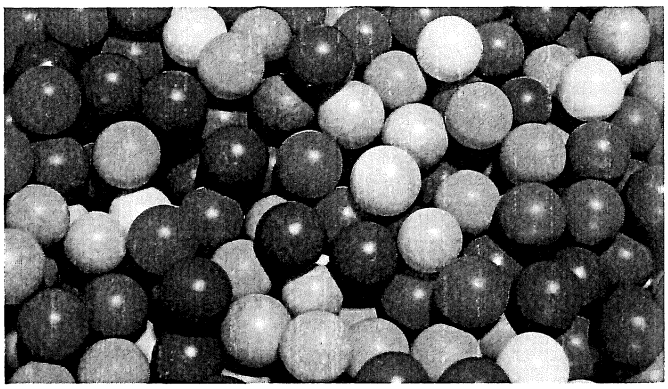


FIGURE 7.1.5 The balls do not change or form new colours despite being spread evenly through this mixture.

)issolving

Vhen a solid is dissolved in a liquid, they form a iixture. The solid is known as the **solute**, the liquid the **solvent**, and the mixture formed is referred to s a **solution**. The smallest particles of the solute mix nd spread evenly throughout the smallest particles of ıe solvent—just like a mixture of balls but invisible ɔ the naked eye. Although the solution looks and acts ifferently to the solute and solvent, no new substance ; formed. For this reason, the process of dissolving is a hysical change.

n a solution, the solute seems to disappear. In reality, he solute is broken down into such small particles hat they are invisible. If the solvent is removed by ɔoiling or evaporation, then the solute particles can join ogether again to form crystals. This process is known .s **crystallisation**.

For example, when sugar is dissolved in water, the ugar particles spread evenly throughout the water o form a sugar solution (Figure 7.1.6). This is a ɔhysical change because the sugar particles remain ugar particles and the water particles remain water ɔarticles—a physical change has occurred. If the water s boiled off or left to evaporate, then the dissolving ɔrocess is reversed and sugar is left behind.



FIGURE 7.1.6 When two substances mix, one substance is spread evenly through the other but no new substances are formed.

Identifying chemical change

To distinguish between a **chemical change** and a physical change, you need to determine whether a new substance has formed as a result of the change. If a new substance has been produced, then a chemical change has occurred.

The new substance formed in a chemical change could be a solid, liquid or gas. You can detect the presence of a new substance by:

- seeing a change in colour
- smelling a gas or seeing bubbles
- seeing a new solid forming in a clear solution. The solid that appears is known as a precipitate
- seeing light or feeling a change in temperature.

These signs of chemical changes are shown in Figure 7.1.7. However, these are just guidelines. Identifying chemical change also requires a good understanding of physical changes because there can be exceptions to these rules.

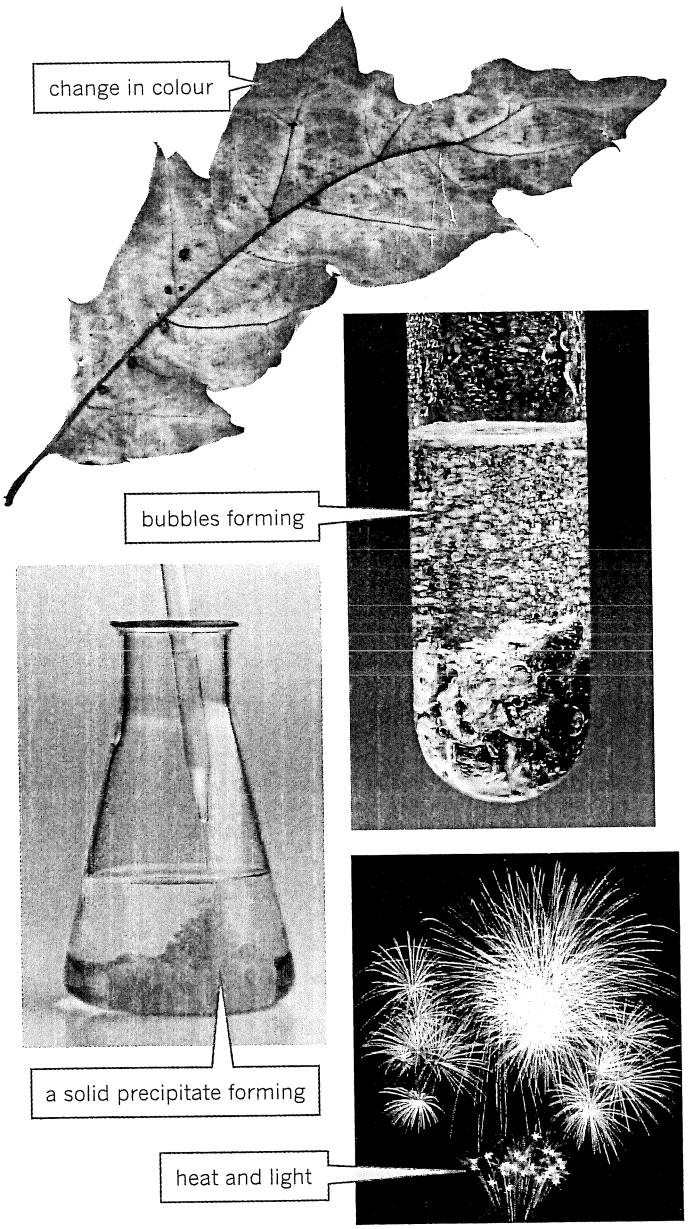


FIGURE 7.1.7 Some indications of a chemical change

ometimes both types of change can happen at
)ughly the same time. For example, toasting bread
:arts as a physical change—heat evaporates water
'om the bread, drying it out. The drying bread then
:arts to burn, changing colour to brown then black.
'his is a chemical change.

\ colour change occurs

: you see something changing colour, then it is very
kely that a chemical change has occurred. This
appens because the new substance produced by the
hemical change has a different colour to what you
tarted with. Burning a match, paper or wood produces
black powder, which is carbon. When iron is left in
ıe rain, it turns orange-brown because iron oxide
rust) is produced.

)ther examples include green apples ripening to turn
ed, the changing colour of leaves in autumn, a bruised
:g turning purple, bleaching or dyeing clothes or hair
Figure 7.1.8).



:IGURE 7.1.8 Dyeing hair is an example of a permanent colour
:hange caused by a chemical reaction.

)ometimes a physical change can also cause a colour
:hange. For example, when snow melts to form liquid
vater it turns from white to colourless. This is not a
:hemical change because snow and liquid water are
)oth made of water (molecular formula H_2O).

Mixing paint is another example of a physical change
:hat causes a colour change. Figure 7.1.9 shows yellow
ınd blue paint being mixed together. They produce
green paint. However, no new substances are formed.
The yellow and blue particles are mixed so closely
:ogether that they give the illusion of green paint.

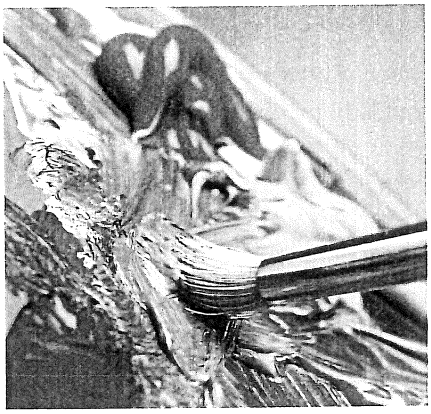


FIGURE 7.1.9
Despite changing colour, mixing paint is still a physical change because no new substance has been formed.

A gas is produced

The production of a gas usually indicates a chemical change. A gas produced during a chemical change may be observed as bubbles, a new smell or perhaps smoke. When vinegar is added to bicarbonate of soda, many bubbles of carbon dioxide (molecular formula CO_2) are formed. Carbon dioxide is also formed by chemical changes in soft drink to create fizz or when antacid tablets are dissolved in water.

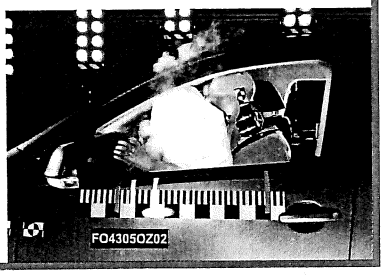
Rotten eggs are well known for their pungent smell. This smell is the gas hydrogen sulfide (molecular formula H_2S), which is produced when eggs rot. Rotting things often produce gas. Therefore, rotting is another example of a chemical change. Often when something is burning, the first thing you notice is the smell. This is because burning causes a chemical change that produces smelly gases.

However, not all gases indicate a chemical change. Sometimes physical changes give off gases too. For example, **boiling** is a physical change in which bubbles of vapour are released from a liquid.

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Life-saving gases

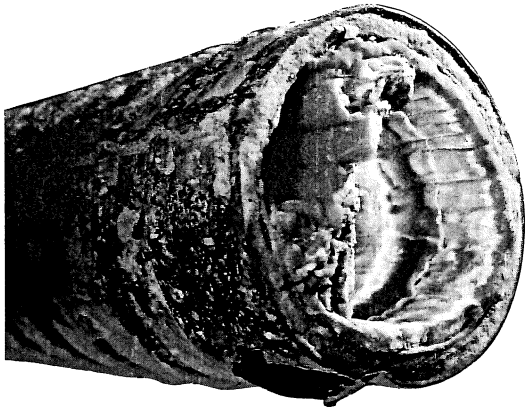
A chemical reaction that has already saved many lives is the one used to inflate car airbags. In a car accident, an electrical current ignites a small pellet of sodium azide. The sodium azide burns very rapidly, producing enough nitrogen gas to fill the airbag in just 0.03 second.



A precipitate forms

Occasionally, a new solid is produced when two solutions are mixed. This indicates that a chemical change has occurred. The solid precipitates (falls) out of the solution. The new solid is known as the **precipitate**. Precipitation occurs in pipes and drains when scale or lime deposits inside, as shown in Figure 7.1.10. Crystals can also precipitate inside the body. Some collect in the kidneys and can cause very painful kidney stones.

A precipitate can also form when a gas is bubbled through a liquid. This type of precipitation is the basis of a simple test for carbon dioxide gas. When carbon dioxide gas is bubbled through a solution of calcium hydroxide (limewater), it precipitates solid calcium carbonate, which turns the clear solution milky white.



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FIGURE 7.1.10 Scale or lime can precipitate from water, causing blockages in pipes.

Energy is produced or absorbed

Energy plays an important role in chemical changes. Some chemical changes release energy in the form of heat, light or both. As a result, a change in temperature or a flash or glow of light can be a sign that a chemical change has occurred. Some chemical changes release huge amounts of energy. For example, the explosion in Figure 7.1.11 released so much energy as light, heat and sound that it would have been observed from far away. Likewise, the light, heat and sound from bushfires, fireworks and sparklers are easily observed. In contrast, some chemical changes give out light but with little or no heat. Examples are the chemical changes that happen in glow sticks and glow worms.

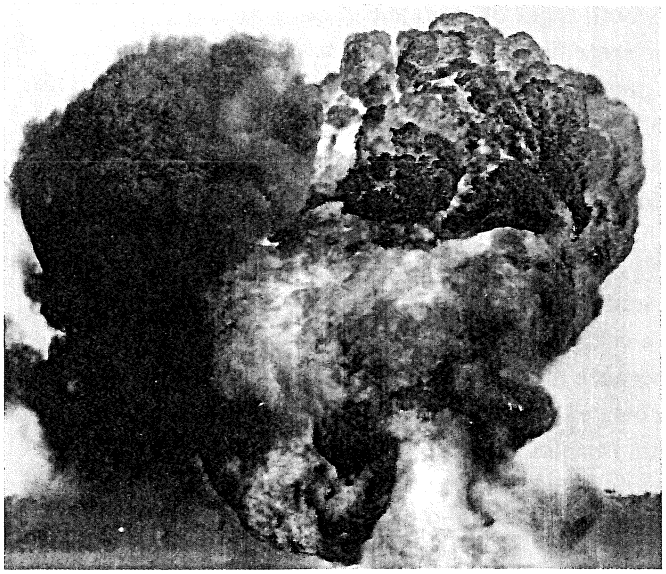


FIGURE 7.1.11 Fires and explosions are examples of extreme exothermic reactions—they release large amounts of heat and light energy.

Other chemical changes absorb heat, causing their surroundings to cool. For example, the chemical ice pack shown in Figure 7.1.12 uses a chemical change that absorbs a lot of heat energy and so feels icy cold. Photosynthesis in plant leaves is another example of a chemical change that absorbs energy, this time from sunlight.

However, some physical changes can also cause a change in temperature. When you sweat, you feel cooler because the moisture evaporates and absorbs some heat from your skin. In contrast, steam can cause serious burns as it condenses on your skin and releases its heat energy.

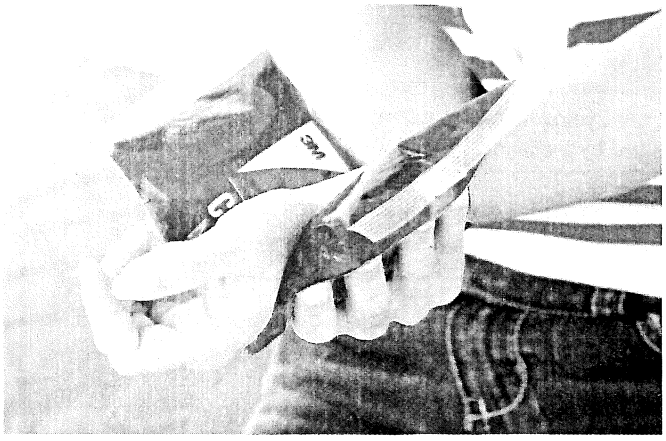


FIGURE 7.1.12 The reaction in a chemical ice pack causes a dramatic drop in temperature as it absorbs heat from the surroundings.

AB
7.2

MODULE 7.1

Review questions

Remembering

1 Define the terms:
   a change of state
   b physical change
   c chemical change
   d precipitate.

2 What term best describes each of the following?
   a A solid is heated to form a liquid.
   b A gas cools to form a liquid.
   c A solute mixes with a solvent and seems to disappear.

3 List four examples of physical changes.

4 List four signs of a chemical change.

5 State which type of change:
   a is easily reversible
   b produces new substances.

6 What is the reverse process for each of the following changes of state?
   a condensation
   b freezing.

7 What are the two components of a solution?

Understanding

8 Mix red and yellow paints and you get orange paint. Explain why this is a physical change even though there is a colour change.

9 Explain how perspiring helps to keep you cool on a hot day.

Applying

10 Identify three examples of both physical and chemical changes you might observe in the home.

11 When a solute is dissolved in a solvent, the solute particles are distributed evenly through the solvent. Identify which of the following diagrams in Figure 7.1.14 best represents what happens when the particles of the blue solute are dissolved in a beaker of water.

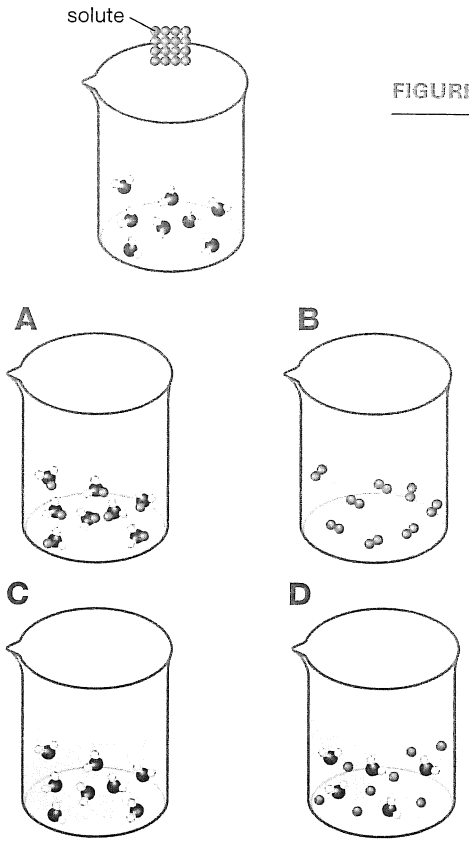


FIGURE 7.1.14

Analysing

12 Classify the following as a physical or chemical change.
   a A banana goes black after a few days.
   b Strawberries are blended with milk to make a smoothie.
   c Wet clothes dry on the clothesline.
   d Spilt petrol explodes.
   e Steel roofing expands on a hot day.
   f Water boils on the stove.
   g A sausage is cooked on the barbeque.

13 A recipe for a banana cake instructs you to carry out the following steps. Classify each step as a physical or chemical change.
   a mash a banana
   b melt butter
   c mix sugar and eggs into butter
   d bake for 40 minutes
   e whip cream for topping.

MODULE 7.1

Review questions

Evaluating

14 You are performing a chemical reaction and notice condensation on the outside of the test-tube.
   a Do you think the reaction absorbed or released heat?
   b Justify your answer.

15 Terry spilt some lemon juice on his homework. When he tried to dry it with a hair dryer, the pages quickly turned brown.
   a Classify the change that Terry observed as a physical or chemical change.
   b Justify your answer.

16 Jo says that an aspirin fizzing and dissolving in water is a physical change (Figure 7.1.15). Joe says that it is a chemical change. Discuss whether Jo or Joe is right, or whether they might both be right.

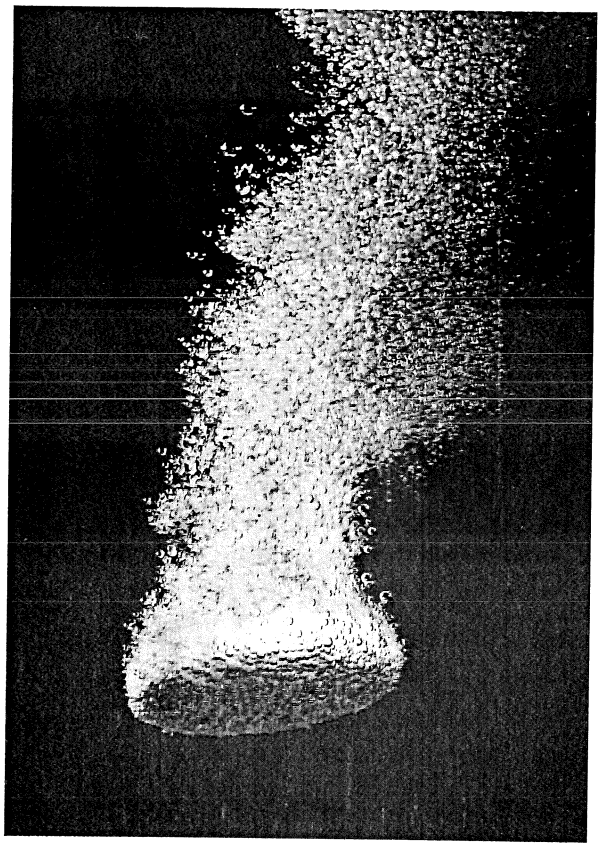


FIGURE 7.1.15

17 How could you reverse the following physical changes?
   a Sand is mixed with water.
   b Salt is dissolved in water.
   c Iron filings are mixed with sand.

18 Draw a table with the following words and define each of the words. Then give an example of materials that can be affected by each of these words.
   a rot
   b crush
   c burn
   d freeze
   e stretch
   f decompose
   g mix
   h rust
   i explode
   j compress.